

Experiment (3)

NEWTON'S LAWS OF LINEAR MOTION

3.1 Introduction

Newton's Laws laid the foundation of classical mechanics, changing people's understanding of the universe at his time. His laws describe the relation between an object and the system of forces acting on it, explaining why and how objects move or not. In this experiment, let us test whether we can verify Newton's Laws in a real life laboratory.

3.2 Objectives

- Investigate the meaning of inertia and experimentally test Newton's First Law of motion. We will use a low friction cart and Cobra4 Sensor- unit Timer/Counter.
- Investigate and verify Newton's second law, which describes the behavior of a particle when a net force acts on it.
- Determine the motion parameters (distance, velocity and acceleration) of a moving body as a function of time, using Cobra4 Sensor- unit Timer/Counter.

3.3 Theory

Newton's First Law of motion states that an object tends to stay in its state as long as there is no net force acting on it. In other words, when the net force acting on an object is zero, if the object is at rest, it tends to stay at rest and if it is in motion, it tends to stay in motion with constant velocity. This means its acceleration (change in motion) is also zero. This tendency of the object to stay in its state is called the '*inertia*' of the object

Newton's Second Law of motion states that as long as a net force F_{net} acts on an object, its velocity changes - in other words, it accelerates. The acceleration of the object is directly proportional to the net force acting on it and is inversely proportional to its mass M , i.e.,

$$\begin{aligned} a &\propto F_{net} \\ a &\propto \frac{1}{M} \\ a &= \frac{F_{net}}{M} \end{aligned} \quad (3.1)$$

Also, the direction of the acceleration is in the same direction as the net force acting on the object.

In this experiment the distance-time law, the velocity-time law, and the relationship between mass, acceleration and force for a low friction-moving cart are determined with the aid of Cobra4 Sensor-unit Timer/Counter.

Consider a low friction cart of mass m_2 which is moving on a horizontal frictionless track. It is also attached to a weight holder of total mass m_1 across a smooth pulley as shown in the Figure 3.1.

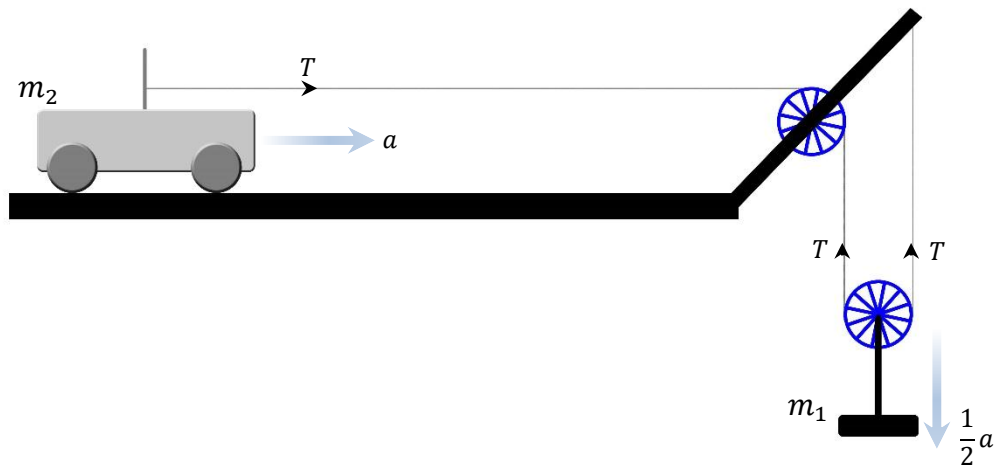


Figure 3.1: A diagram for the experiment setup.

In this case, the motion of the cart and the weights under the force produced by a weight of m_1 , according to Newton's second law is given by:

$$(m_1 + 4m_2) a = 2m_1 g \quad (3.2)$$

where, a is the acceleration of the cart, and g is the gravitational acceleration. So, the acceleration a is given by:

$$a = \frac{2m_1 g}{(m_1 + 4m_2)} \quad (3.3)$$

As can be seen in Equation (3.3), the relation between the acceleration a of the cart mass m_2 and the net external force acting on the system $m_1 g$ is not linear because there is m_1 in the denominator. The equation also shows that the relation between the acceleration a and the reciprocal of the cart mass $1/m_2$ is not linear.

To obtain linear relations between the acceleration a and both $m_1 g$ and $1/m_2$, we may try to keep $m_2 \gg m_1$ through out our experiment. Accordingly, Equation (3.3) can be approximated as follows :

$$a = \frac{\frac{2m_1 g}{4m_2}}{\left(\frac{m_1}{4m_2} + 1\right)} \quad (3.4)$$

but $\frac{m_1}{4m_2} \ll 1$, so $\frac{m_1}{4m_2} + 1 \approx 1$

$$a \approx \frac{\frac{2m_1 g}{4m_2}}{1} \approx \frac{m_1 g}{2m_2} \quad (3.5)$$

3.4 Setup

The setup main parts are shown in Figure 3.2, Figure 3.3 and Figure 3.4.

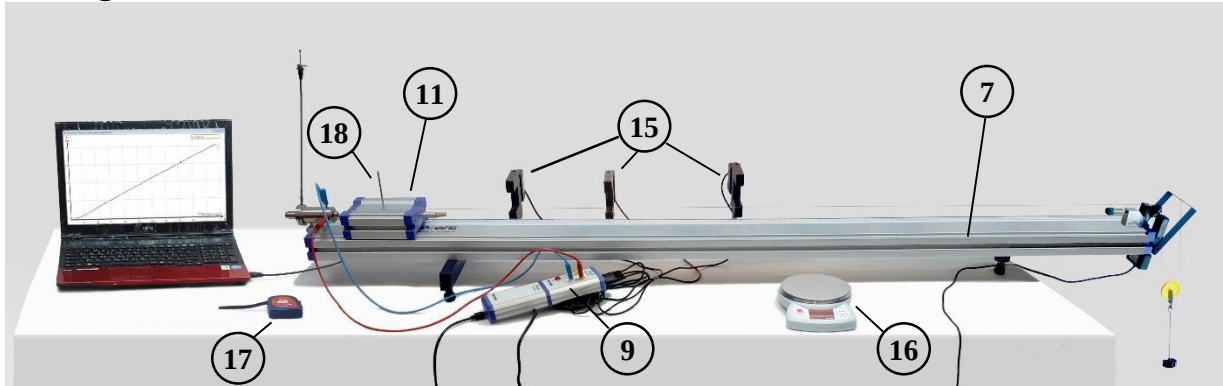


Figure 3.2: Experiment setup.

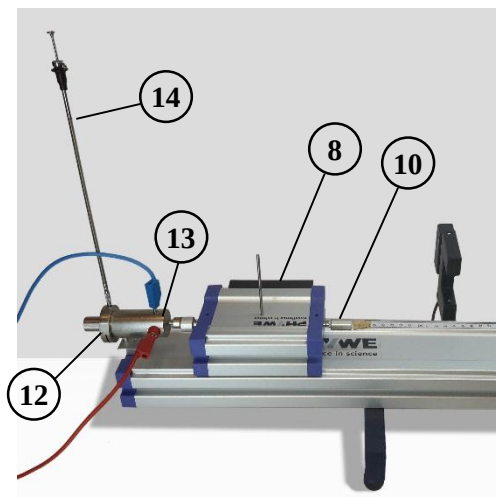


Figure 3.3: The Cart and the starter system.

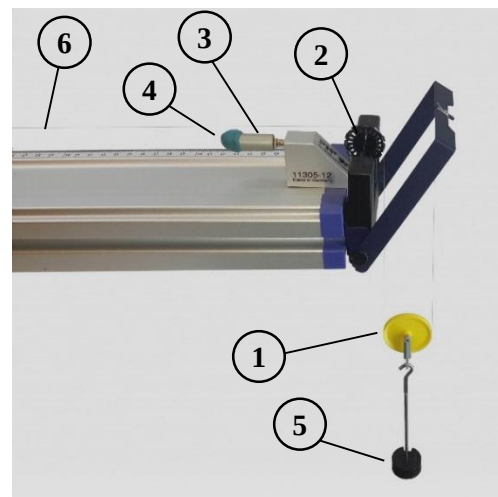


Figure 3.4: The pulley and the wheel incremental encoder.

- 1 Pulley
- 2 Wheel incremental encoder
- 3 Tube with plug
- 4 Plasticine
- 5 Slotted masses and weight holder
- 6 Silk thread
- 7 Demonstration track
- 8 Flat barrier plate
- 9 Cobra4 Sensor- unit Timer/Counter

- 10 Tube with plug
- 11 Cart
- 12 Starter system
- 13 Holding magnet
- 14 Metal plunger
- 15 Light barrier
- 16 Digital balance
- 17 Measuring tape
- 18 Metal pin

3.5 Procedure



A moving cart with a needle is used in the following tasks. **Never put your hands in the path of the cart**, as shown in Figure 3.5.

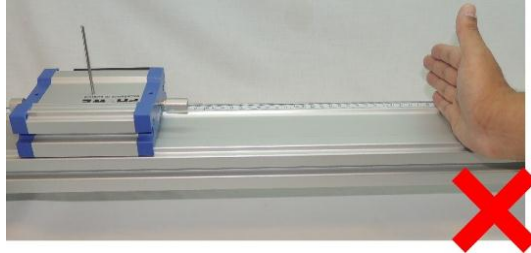


Figure 3.5: Never put your hand in the cart way.

3.5.1 Task (1): Distance–time law, Velocity–time law and acceleration – time law at net force equals zero.

- 1- Adjust the first light barrier (15) to be 20 cm apart from the flat barrier plate using the measuring tape (17) as shown in Figure 3.6.
- 2- Adjust the distance between the light barriers to be 10 cm apart between one another.
- 3- The Cobra4 Timer and counter unit (9) should be connected as shown in Figure 3.7 with the first light barrier from the cart the cart side to be connected at port TC1, the second at 2 and the third at 3.
- 4- Remove the safety plug before pressing the metal plunger as shown in Figure 3.8.

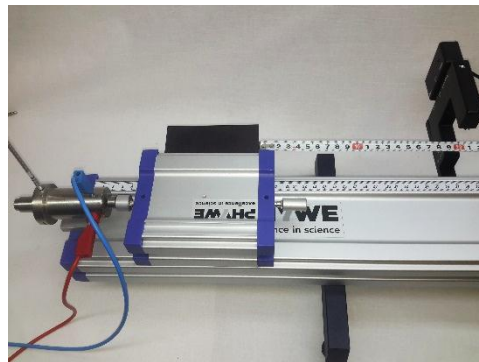


Figure 3.6: Checking the light barriers distance.



Figure 3.7: Connection of light barriers with Cobra4 Timer and counter unit.

- 5- Start the Measure software package on the PC. You will see the window shown in Figure 3.9. Click “Cancel”, then go to the “Experiment” menu and select “Load configuration”.

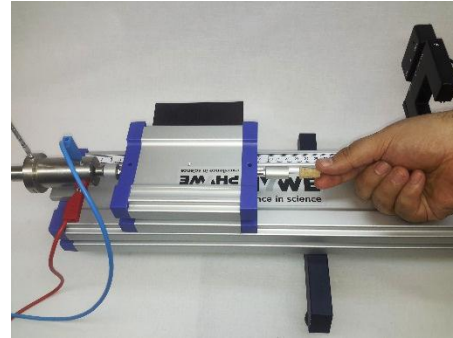


Figure 3.8: Removing the safety plug.

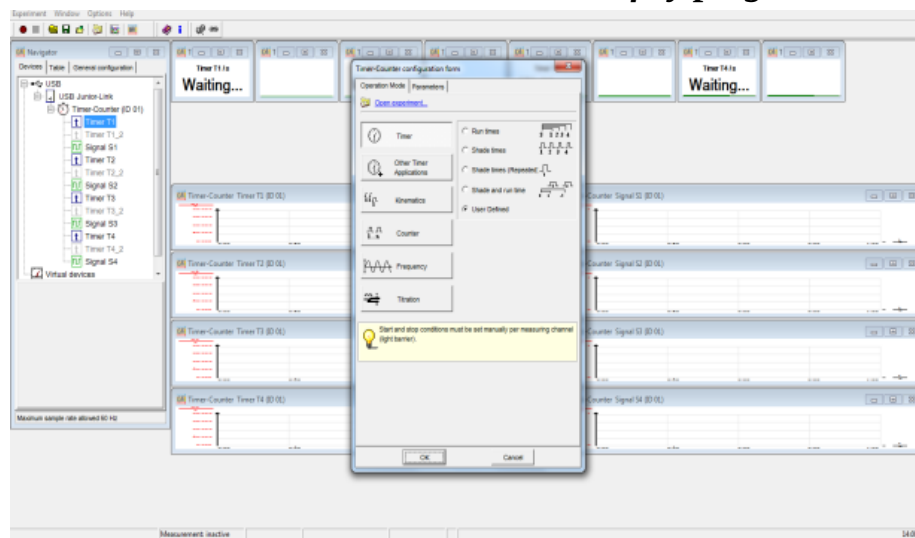


Figure 3.9: Startup window

- 6- From the window shown in Figure 3.10, select “Desktop”, then “Newton’s laws of motion”. Load the configuration file “Newton 1st law”. This configuration file makes the light barriers measure the time at the instant the plate barrier cuts the light beam.

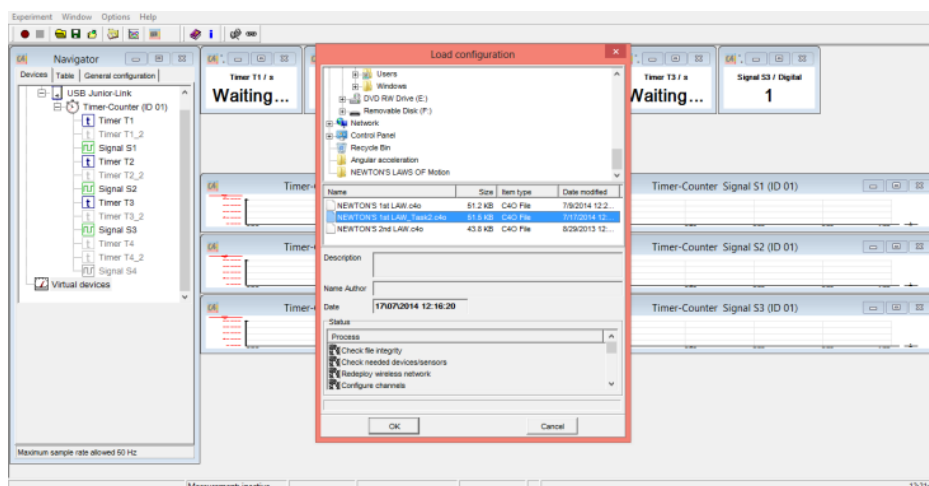


Figure 3.10: Load configuration file.

- 7- After loading the configuration file, a new window will appear as shown in Figure 3.11.

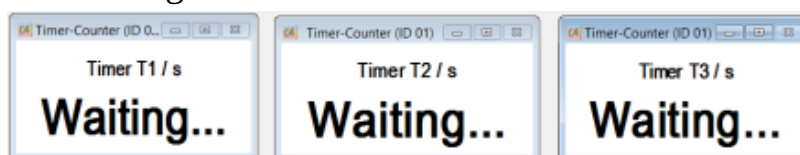


Figure 3.11: Waiting for measuring.

- 8- Adjust the starter system to one of the three defined positions.



The starter system has an internal spring that can be compressed to three defined positions. You can adjust these positions (i.e. by pressing on the holding magnet).

Figure 3.12, Figure 3.13 and Figure 3.14 show the three positions of the starter system. Each of these positions will give the cart different initial velocities.



Figure 3.12: Starter system first position.

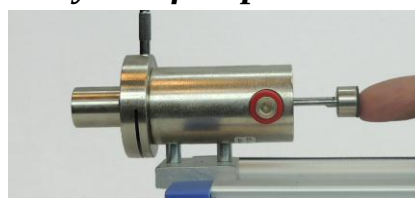


Figure 3.13: Starter system first position.

The maximum initial velocity can be obtained from the third position. You can use any of these positions during this Task but it has to be fixed for all measurments.

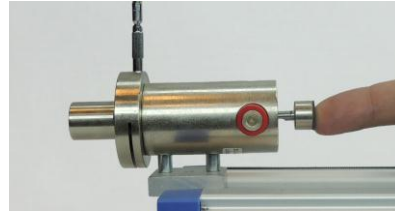


Figure 3.14: Starter system third position.

- 9- Start recording the measured values in the Measure software by prerssing the “Record” button (●).



Figure 3.15: Recording and stop measurments.

- 10- Wait for about 2 seconds then press the metal plunger in the starting device (12) as shown in Figure 3.16 to release the cart and give it an initial velocity.



Avoid mistakenly pressing the plunger without preparing your Measure software. If you accidentally trigger the movement prematurely, you will have to repeat the step.

- 11- Stop the measurement by pressing the “Stop” button (■) after the cart (11) hits the tube with plug (10).
- 12- The data will appear as shown in Figure 3.17.
- 13- Write the measurements in your lab report and plot a graph for the Distance-Time and Average velocity-Time.

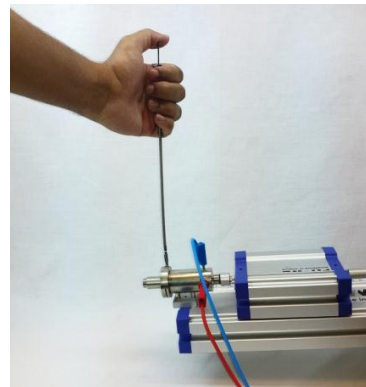


Figure 3.16: Pressing of metal plunger.

Calculate the average velocity by using $v_{avg} = \frac{\Delta s}{\Delta t}$. Find the acceleration using the slope of the Average velocity-Time plot.

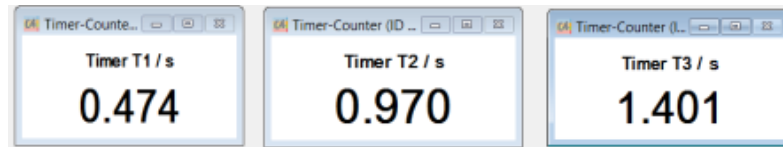


Figure 3.17: Measured values.

14- Now we will calculate the velocity of the cart by another way.

We will use the same setup as shown in Figure 3.2.

15- From the window shown in Figure 3.10 above, select “Desktop”, then “Newton’s laws of motion”, load the configuration file “Newton’s 1st law-Task2”. This configuration file changes the function of the light barriers to measure the time during which the flat barrier plate passes through the light beam.



The light barrier now measures the time by which the flat barrier attached to the car passes through it. The velocity of the cart can be determined using the known length of flat barrier plate = 100 mm. Figure 3.18 shows the instant at which the flat barrier plate starts cutting the light barrier beam. At this time the counter starts counting. Figure 3.19 shows the instant in which the flat barrier plate has completely passed the light barrier beam. At this time counter finishes counting. We divide Time by the length of the flat barrier plate to calculate the velocity.

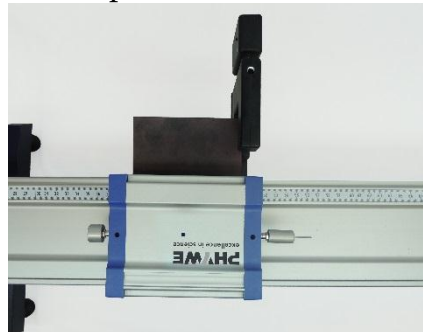


Figure 3.18: the flat barrier plate just cut the light beam (counter starts).

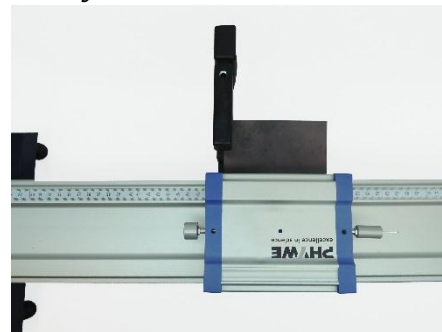


Figure 3.19: the flat barrier plate just passed the light beam (counter stops).

- 16- Repeat steps 8 to 11.
- 17- The data that will appear is the time Δt the flat barrier plate takes to pass the light barrier.
- 18- You can measure the instantaneous velocity of the cart at each light barrier by dividing the length of the flat barrier by Δt .
- 19- Compare the cart velocities by the two methods in your lab report.

3.5.2 Task (2): Determination of the motion parameters (displacement, velocity and acceleration) as a function of time using Cobra4 Sensor- unit Timer/Counter by applying constant force.

- 1- Use the same setup as shown in Figure 3.2 above but connect the Cobra4 unit as shown in Figure 3.20. Connect only the wheel incremental encoder (2) to Cobra4 Sensor-unit Timer/Counter.
- 2- From the window shown in Figure 3.10, select “Desktop”, then “Newton’s laws of motion”, load the configuration file “Newton 2nd law”.
- 3- Connect the metal pin in the cart as shown in Figure 3.21.



Figure 3.20: Connecting the wheel incremental encoder to Cobra4 Timer and counter.

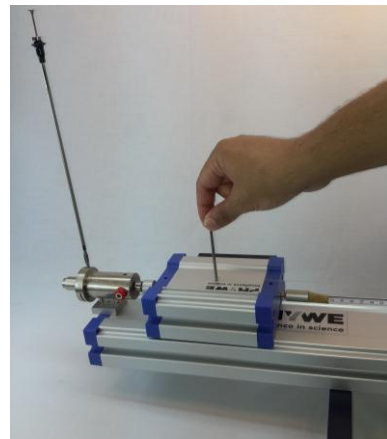


Figure 3.21: Connecting the metal pin.

- 4- Remove the wires from the metal plunger as shown in Figure 3.22.
- 5- Rotate the starter system to the new position as shown in Figure 3.23 and Figure 3.24.



Figure 3.22:
Removing the wires.

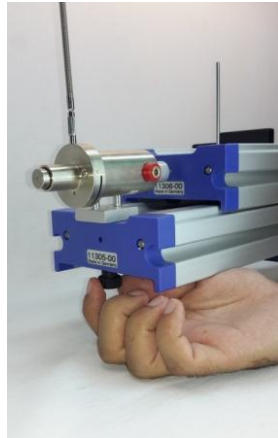


Figure 3.23:
Loosing the screw.



Figure 3.24:
Adjusting the starter system.



Make sure that the movable hook is in a position such that the two sections of the silk thread that hold up the mass and lower pulley are both vertical (parallel to each other) as shown in Figure 3.25.

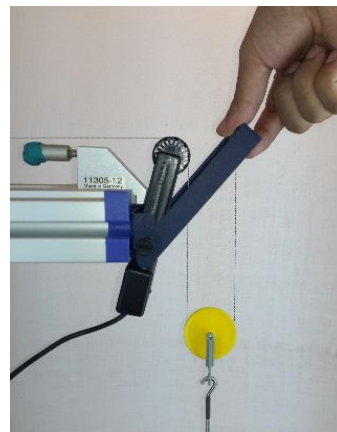


Figure 3.25: *Segments of the silk thread parallel to each other.*

- 6- Use the weight holder without any extra added masses (Empty weight holder mass = 10 gm).



The starter system in this position holds the cart by attaching it to the holding magnet as shown in Figure 3.26 and Figure 3.27.

When the metal plunger is pressed, the holding magnet is no longer attached to the cart, allowing the cart to move under the force of the weight holder as shown in Figure 3.28.

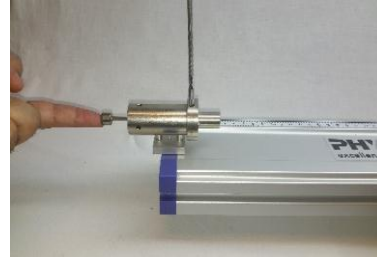


Figure 3.26: Adjusting the holding magnet.

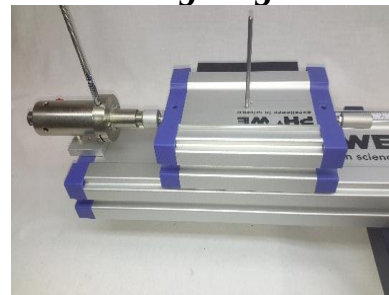


Figure 3.27: Attaching the cart to the holding magnet.

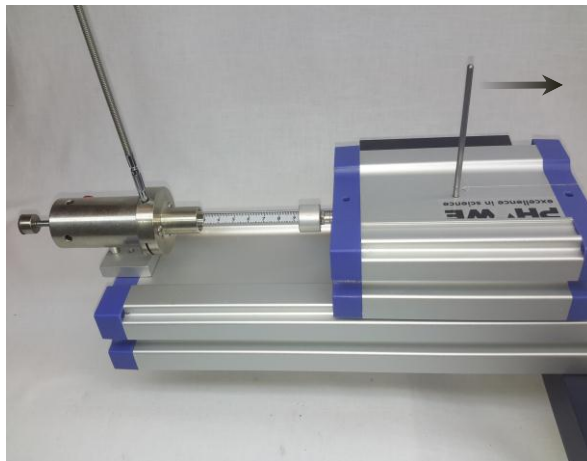


Figure 3.28: The cart moves under the applied force from weight holder After pressing on the metal plunger.

- 7- Start recording the measured values in the Measure software by pressing the “Record” button (●). At the same instant, press the metal plunger in the starting device (12) as shown in Figure 3.16 above to release the cart.

- 8- Stop the measurement by pressing on () after the cart (11) hits the tube with plug (10).
- 9- Transfer the measured data to the Measure module by pressing “send all data to measure”.
- 10- A new window will appear as shown in Figure 3.29.

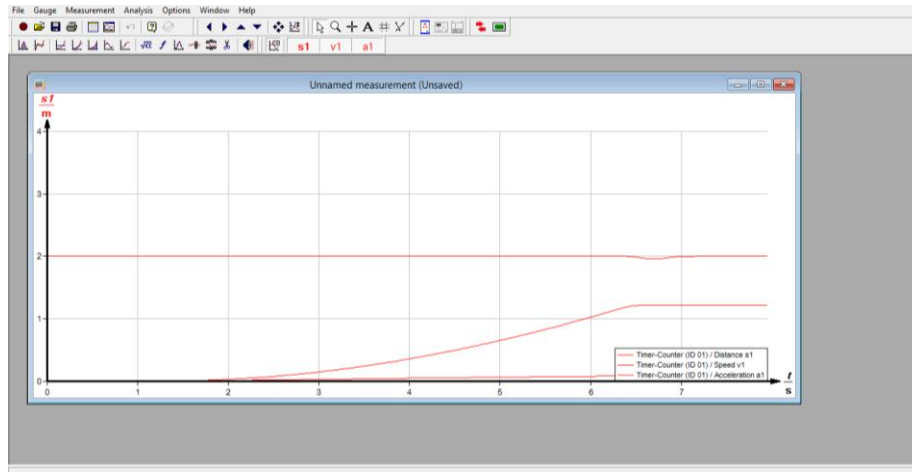


Figure 3.29: Distance – Time, Velocity-Time and Acceleration-Time curves.

- 11- Study the curves of Distance – Time, Velocity – Time and Acceleration – Time for an accelerated cart by pressing on () and unpress both () , ().
- 12- You will see the Velocity-Time curve and Acceleration-Time curve as shown in Figure 3.30 and Figure 3.31.

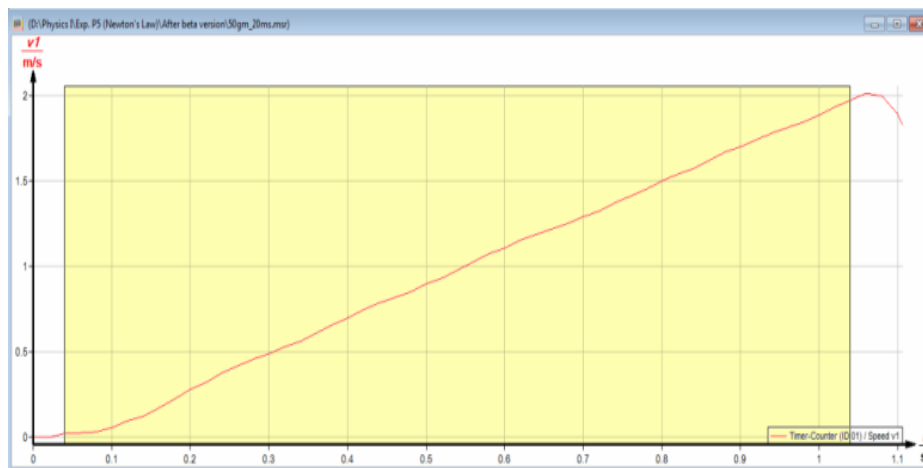


Figure 3.30: Velocity-Time curve.

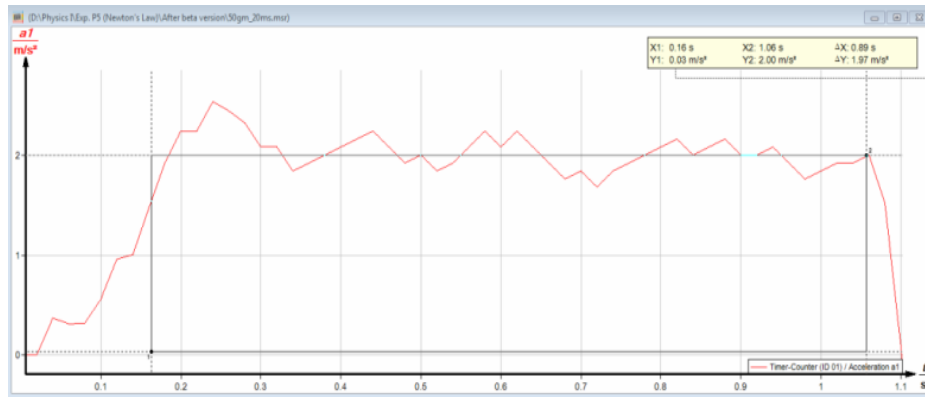


Figure 3.31: Acceleration-Time curve.

3.5.3 Task (3): Relation between the acceleration and the mass of the body at constant net external force.

- 1- Use the same setup as in Task (2) with the same configuration file.



Before every reading, make sure that the plasticine (4) is not flat so it can absorb the impact of the cart as shown in Figure 3.32. Make sure that the silk thread (6) is wound around the wheel of the light barrier (2) as shown in Figure 3.33 and that the pulley (1) is set as shown in Figure 3.34 and that the magnet (13) is adjusted to hold the cart as shown in Figure 3.35.

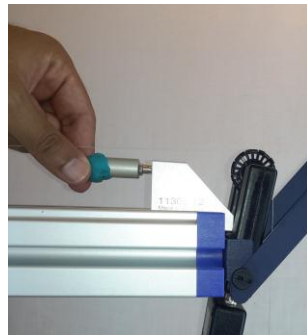


Figure 3.32: Adjust the plasticine.



Figure 3.33: Put the silk thread around the wheel.



Figure 3.34: Put the silk thread around the pulley.

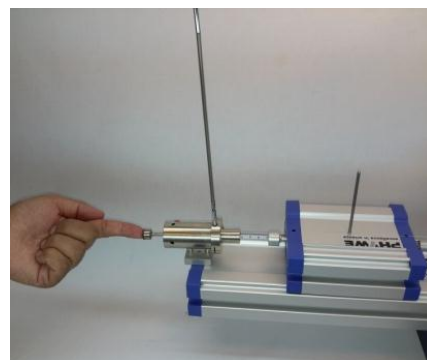


Figure 3.35: Adjust the magnet to hold the car.

- 2- Add 10 gm slotted mass on the cart and remove any added masses from the weight holder (Empty weight holder mass = 10 gm).
- 3- For light added masses, you may set the “Sampling time” to 20 ms as shown in Figure 3.36 and for heavy added masses you may set this time to 50 ms.
- 4- Repeat steps 7 to 10 from Task (2).
- 5- Select the velocity curve by pressing (**v1**) and unpress both (**a1**) and (**s1**).
- 6- Use the “Regression Tool” () to make a straight line, which conforms to the relationship $v = at$. A regression line will fit the measured points but to consider only the linear part of the curve, place one of the blue points of the regression line at the beginning of the linear part of the curve and the other blue point at the end of the linear part of the curve as shown in Figure 3.37.

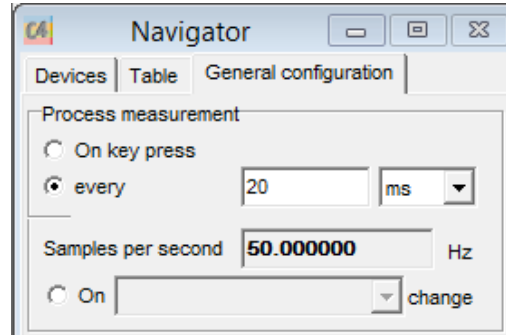


Figure 3.36: Adjusting the sampling time.

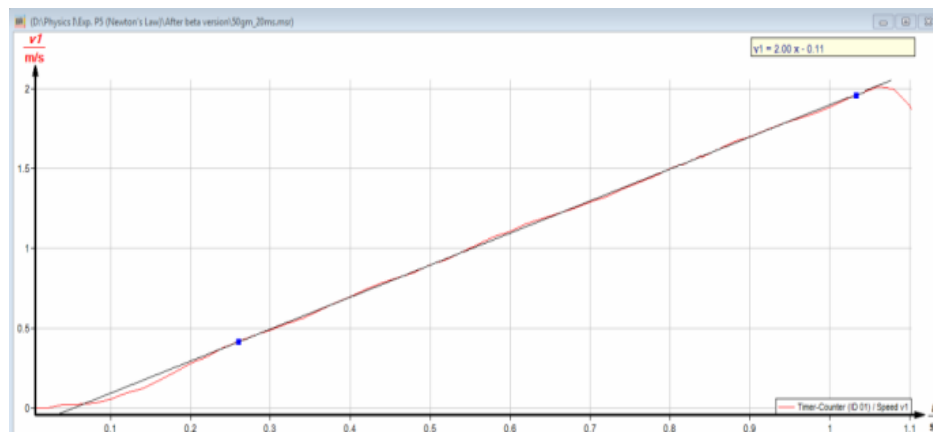


Figure 3.37: Velocity-Time curve with linear regression curve.

- 7- Select the acceleration curve by pressing (**a1**) and unpress both (**v1**) and (**s1**). Use the “Survey Tool” () with Acceleration-Time curve to determine the average value of acceleration as shown in Figure 3.38.



Figure 3.38: Acceleration-Time curve.

- 8- Repeat steps 4 to 7 another four times and each time add another 10 gm to the cart.
- 9- Measure the mass of the cart without any added masses using the digital balance (16).
- 10- Plot the relation between the acceleration a of the cart and the reciprocal of its total mass (cart mass and the added masses) $\frac{1}{m_2}$ at constant net external force acting on the system.

3.5.4 Task (4): The relation between the acceleration of a coupled system and the net external force acting on it.



For the coupled system, we can treat the two masses, the pulley and string as a single system then the acceleration of the cart and the acceleration of the weight holder are related by a constant factor and the net external force on the system is m_1g .

- 1- Use the same setup and configuration file as for Task (2) and (3).
- 2- Remove any added masses from the cart.
- 3- Use the weight holder without any extra added masses (Empty weight holder mass = 10 gm).
- 4- Repeat steps 7 to 10 from Task (2).
- 5- Repeat steps 5 to 7 from Task (3).
- 6- Repeat steps 4 to 5 another 5 times and each time add 10 gm slotted mass to the weight holder to have total hanging masses as follows : 20 gm, 30 gm, 40 gm, 50 gm and 60 gm.
- 7- For heavy weight holder, you may set the "Sampling time" to 20 ms as shown in Figure 3.36 above and for light weights you may set this time to 50 ms.
- 8- Plot the relation between the acceleration a and the net external acting force m_1g at constant cart total mass.